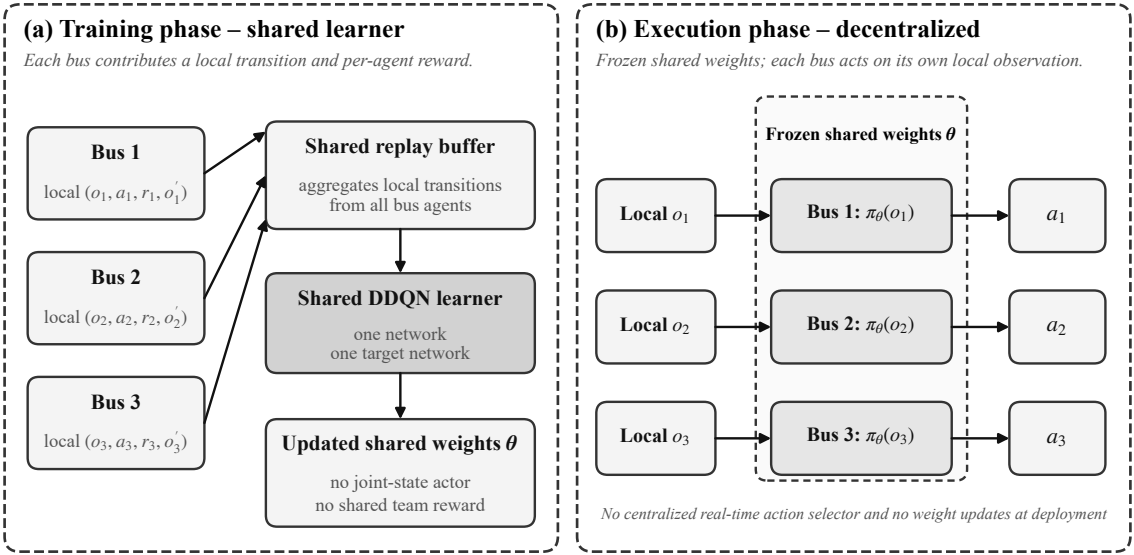

(a) Training phase – shared learner
Each bus contributes a local transition and per-agent reward.
Bus 1
local $(o_1, a_1, r_1, o_1')$
Bus 2
local $(o_2, a_2, r_2, o_2')$
Bus 3
local $(o_3, a_3, r_3, o_3')$
Shared replay buffer
aggregates local transitions from all bus agents
Shared DDQN learner
one network
one target network
Updated shared weights $\theta$
no joint-state actor
no shared team reward
(b) Execution phase – decentralized
Frozen shared weights $\theta$
Local $o_1$
Bus 1: $\pi_\theta(o_1)$
$a_1$
Local $o_2$
Bus 2: $\pi_\theta(o_2)$
$a_2$
Local $o_3$
Bus 3: $\pi_\theta(o_3)$
$a_3$
No centralized real-time action selector and no weight updates at deployment